

Confucius Tool Verification Paper

Abstract

This deterministic paper exists solely to verify Zotero library, full-text, section, page, search, selection, and annotation tools. Its wording is stable so automated assertions can distinguish a real result from a guessed answer.

Confucius verifies real PDF highlighting.

1 Introduction

A reliable tool test needs a known document, a known phrase, and visible evidence in the reader. The token ALPHA-VERIFY-2026 appears only on page one. Searching for that token should return exactly this context.

The first page also establishes the expected author-independent metadata: a title, a 2026 date, and a synthetic DOI stored on the parent Zotero item.

2 Methodology

The verification sequence creates a collection and two bibliographic items, links them as related, writes and edits a child note, attaches this PDF, and then exercises every read tool against the resulting keys.

The exact phrase BETA-RECT-4242 is reserved for page-two text selection and highlight geometry checks. A genuine PDF highlight must appear over this text inside the Zotero reader; creating only a child note does not satisfy the test.

3 Results

A passing implementation reports structured tool results, persists intended writes, leaves rejected writes untouched, and exposes the same state through Zotero's visible interface. The matrix records pass, fail, and blocked outcomes.

4 Discussion

Fallback behavior must be labeled honestly. If annotation commit produces a note instead of a reader annotation, the tool transport works but PDF highlighting remains functionally incomplete.

The token GAMMA-FINAL-9000 appears on the final page for range and regex tests.

5 Conclusion

This fixture is intentionally small, searchable, and visually plain enough that both code assertions and Computer Use can verify the same facts.